Results of the 9th Scientific Workshop of the European Crohn's and Colitis Organisation (ECCO): Artificial Intelligence in IBD: Regulatory and Methodological Considerations

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Abstract

With the rapid growth of artificial intelligence (AI) applications in the field of inflammatory bowel disease (IBD), an increasing number of regulatory and methodological considerations have become apparent. Currently, there remains much uncertainty and limited experience in the field of IBD regarding some of the regulatory and methodological pitfalls to be considered when developing and deploying AI applications for positive clinical and health system impact. Accordingly, an expert panel was convened by the European Crohn's and Colitis Organisation (ECCO) to review the published literature and provide an overview of key regulatory aspects for the application of AI in IBD. This article discusses and, where possible, provides guidance on key methodological and regulatory considerations for AI in IBD. Topics covered include: potential clinical application-focused algorithm design; ethical, moral and legal considerations; regulatory agency perspectives; an overview of regulatory submission and consideration of reimbursement. By providing clinicians with a primer to key regulatory and methodological considerations, we hope to accelerate knowledge translation and implementation of AI-enabled digital health innovations in clinical practice and ultimately improve outcomes for people living with and caring for those living with IBD.

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Introduction

Patients with inflammatory bowel disease (IBD), including the two main subtypes of Crohn's disease (CD) and ulcerative colitis (UC),[1,2] can present with a wide range of symptoms due to the systemic nature of IBD.[3,4] A multidisciplinary and multiprofessional team of healthcare providers typically collect a wide array of information before eventually arriving at a diagnosis. A shared decision-making process is then used to help patients select therapy guided by medical knowledge, clinical experience and professional treatment guidelines.[5–8] As patients progress through their disease course, further decisions pertaining to management are critically revisited and adjusted. Over time, IBD expert clinicians (and patients) learn from outcomes and increasingly gather experience that informs their future decisions.[9] However, due to the exponential growth of medical knowledge, this incremental knowledge-gathering approach to diagnosis, outcomes and treatment can no longer be considered sufficient. Clearly, more innovative methods are required to process the enormous volume of information so that it can be translated into therapeutic success, innovative discovery and health economic performance in IBD care—particularly given the rapidly growing economic burden of disease[10] and the increasing number and complexity of available treatment options.[11]

The digital revolution in medicine began some 15 years ago.[12] Advancements in information and communication technology—including mobile medical devices[13] and wearables,[14] the availability and sharing of electronic health data at scale,[15] powerful computer hardware designs that overcome the traditional computing architectures with separation of memory and processing units[16] and artificial intelligence (AI) methods,[17–21]—have helped to accelerate, enhance and refine medical knowledge acquisition and

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translation, a process that promises to transform biomedical research, medicine and healthcare delivery.[22]

AI, although widely accepted to be a disruptive advance in health technology, does come with new potential risks, ethical and moral dilemmas[23] as well as bias and health disinformation,[24] and in some cases it can even result in harm.[25] Therefore, AI applications need to be careful weighed, with consideration of benefits versus risks.[26] Balancing benefit and risk in medicine is perhaps one of the most daunting tasks and rightfully requires a well-designed, structured regulatory approval process. Indeed, this helps explain why the regulatory process requires consideration of the evidence regarding benefit and risk even before any regulatory application is filed. In the face of new technologies, regulatory bodies have had to significantly adapt their guidance and workflows in order to strike a balance between minimisation of potential risks and maximisation of potential benefit from AI applications.[27,28]

The US Food and Drug Administration (FDA) has opted to regulate AI applications as a device (further details below).[29] This is in line with the International Medical Device Regulators Forum (IMDRF) commitment to develop a harmonised approach to the management of AI as part of their 2021–2025 strategic plan.[30] In October 2023 the entire G7 group of nations, i.e. Canada, France, Germany, Italy, Japan, the United Kingdom and the USA, as well the wider EU, passed an agreement on International Guiding Principles on AI[31] and a voluntary Code of Conduct[32] for AI developers, referred to as the Hiroshima AI process. This was done to ensure that patient outcomes would be prioritised throughout regulatory processes.[33] In particular, the Hiroshima AI process focused on the importance of reproducibility,[34] transparency

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(explainable AI, commonly referred to as “XAI”), decision autonomy level, decision effect size and opportunity for human oversight and override as critical factors for the approval of any AI application. It has especially been recognised that post-marketing surveillance and liability[35] for approved AI applications are particularly complex, given that industrial AI models, unlike drug approvals, are not static over their lifecycle.[36,37]

Here we aim to give an overview of the creator journey from an idea to regulatory submission, with particular focus on AI tools aimed at improving outcomes for patients living with IBD (**Figure 1**).

Methods

ECCO commissioned a Scientific Workshop Consensus Group on the subject of AI in IBD. Following an open call across all ECCO Members, 18 participants were selected based on their expertise, and four subgroups were formed. This article highlights the topic addressed by Working Group 4: regulatory and methodological aspects of AI in IBD.

The working group performed a comprehensive literature search of the following databases: MEDLINE, Embase and Cochrane CENTRAL via Ovid on 11 July 2024. Additional references were identified through reviewed articles. The search terms and search strategy are provided along with the PRISMA flow diagram (**Supplementary Figure 1**). In several areas covered within this manuscript, it was recognised that the quality of evidence was low, which reflects the paucity of research/literature, including a lack of randomised controlled trials (RCTs) for AI applications in the context of IBD.

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Accordingly, where appropriate, expert opinion was included and translatable examples from other areas of medicine and healthcare were used.

Envisioned clinical setting and jurisdiction of AI application

AI tools could potentially be applied in any IBD setting where there is a requirement to synthesise data from multiple sources, determine an outcome and then select from a range of options in order to maximise benefit for patients. In this regard, all aspects of diagnosis, prognosis, treatment selection, monitoring and even prevention could, in principle, be informed by AI tools and models.[38,39] However, there are several ethical, moral and legal aspects of AI to consider when developing a potential tool for application in IBD.

In terms of ethical and moral considerations, a critical aspect of AI development relies on the principle of fairness and seeking to avoid bias. Moreover, there are profound cultural differences in the perception and expectations of technology. Opinions on what AI could or should do in challenging scenarios are highly diverse and a source of controversy. This variation exists not only between advocates and opponents, but also among different generations and individuals from distinct geographical locations. Indeed, such ethical and moral divergence between individuals was clearly demonstrated in the Moral Machine experiment, where an online experimental platform demonstrated that decision-making between individuals was highly variable in the face of moral and ethical dilemmas.[40]

AI systems, depending on the datasets on which they are trained and validated, may perpetuate or amplify known or unknown biases. A well-recognised problem from early

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AI development was recognition that tools are often trained on highly selected cohorts, which can compound inequities in race, ethnicity, gender, socioeconomic status and many other factors.[41] Given that there is a need for AI tools to be applicable, to be accessible and to benefit diverse populations of patients,[42] the datasets on which AI tools are trained should ideally be large and themselves composed of diverse patient populations. Therefore, a key task in AI development now and in the future is to determine how best to evaluate for these biases and develop solutions to address biases when they do occur.[43]

One major problem which can arise from seeking to develop and validate AI tools in very large datasets relates to the data itself. It is widely recognised that diversity and high quality of patient data in a training dataset result in a greater likelihood of true population representation in the resulting AI model. However, this raises concerns about infringement of personal rights and how appropriate consent will have been obtained from individuals to allow data generated from their healthcare encounters to be used in such a way. Alternative methods have been proposed, such as generation of synthetic data,[44] but these have been counterbalanced with concerns that such measures may decrease the performance and clinical utility of AI tools. Moreover, use of generalised and historical data needs to be weighed against the increasing aspiration of patients and families for personalised care. One thing is clear: safeguarding rights and privacy for individuals is an important but challenging topic, with additional complexity deriving from differences between legal systems and the ways in which data can be handled and transferred between distinct legal jurisdictions. Although still some way off implementation, there are potential solutions, including

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depersonalisation of data, with separation of medical/healthcare-related data from individual person-related demographic data.

Ethical algorithm design is covered in more detail later in this manuscript. However, it is clear that AI tools which operate as “black boxes” are likely to encounter difficulty in being successfully implemented into routine clinical care.[45] “Black box AI” refers to tools where it is not clear to regulators, clinicians and patients how data is being processed and how decisions are being made. Indeed, a crucial part of shared decision-making by patients with their healthcare providers is an appreciation of the tangible benefits and risks from different decisions. A black box outcome where it is not clear how those benefits and risks have been handled removes autonomy from both clinicians and patients. This lack of transparency alone makes it hard to reconcile with current clinical practices and long-term care of chronically ill patient populations such as the IBD population.

One further ethical, moral and legal problem that has likely limited uptake of AI tools has been concern about what to do if an AI tool provides the “wrong” answer. This could include scenarios where an AI tool guides patients and clinicians down a clinical management pathway that does not improve patient outcomes or even results in harm to patients. Such instances are a reminder that responsibility and accountability for decisions, even when guided by decision-support tools, still lies with the healthcare provider. An obvious lack of accountability arises from the use of unregulated and unapproved AI healthcare tools. One such example would be to feed patient data into an internet-based version of the LLM ChatGPT (Microsoft, Redmond, WA), which would likely cause much unease amongst fellow clinicians and patients alike.[46] Legal

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responsibility resides with healthcare providers and the health systems in which they practice. One potential downside of this associated legal responsibility is that some individuals may be unwilling to adopt new AI-based technologies, no matter how promising the tool may be for improving patient outcomes.

The focus should be on education, training and reconciliation-based approaches to aid learning and further development of AI, rather than on liability and litigation. This, accompanied by financial redress where appropriate, may help increase the adoption of AI tools in the field.[46] Such a “loop-in” approach focuses on adoption across a system. However, this requires individuals to feel sufficiently well informed by decision support systems[47,48] that they have adequate confidence in an AI tool to use the output from that tool to make a clinical decision.

Another ethical and moral dilemma is the inequality of AI healthcare provision and concerns about the availability of AI tools in healthcare settings around the world.[49] If the cost or means of implementing AI tools results in them being used only in high-income countries or only in specialist centres, then this in itself will mean that large numbers of patients do not have access to or the opportunity to benefit from such tools. Conversely, in healthcare settings where there are minimal resources, it might be argued that funding is more appropriately spent on the provision of basic medical care rather than on AI tools, which may be of less clinical utility and benefit for patients.

Just as there needs to be diversity of populations on which AI tools are trained, there equally needs to be consideration of how to make AI tools accessible to all those who may benefit from their use in clinical care. This is a major challenge, and one which

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has been faced by companies developing medications, medical devices and many other biomedical and healthcare interventions over the years. Potential solutions include early adoption and reimbursement incentives to provide access for lower- and middle-income populations and countries, continued access to AI tools following participation in research studies and philanthropic donation of AI tools based on revenue generated from reimbursement and sale in high-income countries.

A critical point, touched upon earlier, relates to the source data used to train and develop AI tools for potential implementation in IBD. In addition to the requirement that source data are obtained from cohorts of diverse patients, there is a need for access to data of a quality and depth that will allow the development of sufficiently robust AI tools. An important potential application of AI is the use of computer vision to provide quantitative assessment of disease extent, disease complications and therapeutic response from various imaging modalities such as video-recorded endoscopy, ultrasound, computed tomography and magnetic resonance imaging.[50] To unleash the potential of such applications, harmonisation of image acquisition, exam documentation and archiving needs to be agreed.[51] We believe that professional societies in the field of IBD can help support the development of frameworks for such efforts.

Ethical algorithm design

AI algorithms, for instance those based on convolutional neural networks (CNNs), have traditionally led to relevant results in replicating human behaviours. However, data weighting and any updates, as well as the handling of biases and the ways in which datasets are used to generate such models, have all typically been considered

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in a black box approach, where only the final output has been evident to the human user.

It has become apparent that as the use of AI has grown across sensitive areas, including health and biomedical sciences and finance, users have increasingly desired to know and understand how the process of algorithm creation considers individual variables. This is perhaps unsurprising in healthcare, where clinicians and patients have sought to obtain this information so that they can appreciate the nuances and details in order to better inform shared decision-making about patient management. As a result, there has been a move towards more transparent AI models and a focus and design shift towards XAI. Explainable AI details the instruments and tools used in a readily comprehensible way, allowing users to achieve a greater understanding of how an outcome has been generated and the data on which it has been based.[52]

As AI models become more pivotal in guiding clinical decision-making, the ethical, moral and legal implications will become more apparent. In a future in which an AI model may strongly recommend prescription of a certain medication for a patient with IBD, it would be of much interest to ascertain how individual patient, disease and treatment features have been considered. Indeed, given the complexity of decision-making and evidence synthesis, patients' and clinicians' views may differ significantly based on the same data. Moreover, AI models (like human beings) will often come to a decision even in the absence of data or on the basis of only a paucity of information. Therefore, awareness of how a model generates decisions/recommendations and the uncertainty behind those decisions/recommendations will likely be crucial for the final user(s) of any AI model.

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It is crucial to understand the implications of uncertainty in AI predictions in order to enable their adoption in clinical practice. Recent examples include a prediction model and treatment algorithm generated from chest X-ray images from patients with coronavirus disease 2019 (COVID-19), where the uncertainty of the model was disclosed to clinicians as a relevant part of their decision-making process.[53] This example demonstrated that CNN prediction may be very accurate. However, confidence in the tool was supported by availability of at least an estimate of the uncertainty; this helped healthcare professionals to better interpret results and reach informed decisions from AI outputs. The opposite would be the case with classical algorithm outputs using a black box approach, where clinicians would have no knowledge of the details of the AI process. Given the high level of trust and the lower level of control/understanding of AI models using a black box approach, it is unlikely that such AI models would be widely adopted in routine clinical care.

In recent years, distinct (technical) approaches have been proposed to support more XAI. Examples include local interpretable model-agnostic explanations (LIME) and SHapley additive exPlanations (SHAP), which are the most commonly used methods and which lead to similar XAI results, along with more structured approaches such as class activation mapping (CAM), gradient-weighted class activation mapping (Grad-CAM), counterfactual explanations and anchors. However, even these technical strategies cannot completely explain the majority of CNNs and model algorithms. Indeed, one major challenge that the field has yet to address is how to make models fully explainable in the face of increasing complexity of the AI algorithms being developed.

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A major challenge in AI models, as highlighted above, is the dataset used to train and test any given model. This issue becomes even more relevant if we explore the application of XAI models to clinical decision-making for IBD treatments. If such models are developed from RCTs, a major issue is that in most cases the majority of trial participants are not representative of “real-world” clinical cohorts with IBD.[54] Accordingly, provision of precise information particularly relating to populations used in the original dataset employed to develop AI models should be part of any XAI process in the context of IBD.

One further problem is that the more AI models become explainable and transparent, the more patenting and intellectual property rights can potentially be at risk. The developers of a hypothetical original AI model supporting medical decisions could be disincentivised to disclose too much information fundamental to the development of their algorithms, which if revealed might result in patent infringement or replication of their models by competitors. These conflicting interests and risks should be carefully analysed, weighed and managed by regulatory bodies in order to avoid the paradoxical drawback that XAI may limit further expansion of newer XAI-aware applications in the field.

We envision a balance between proprietary components (“black box”) and more detail about algorithm design in healthcare applications (allowing physicians to know details of the datasets used and the weights applied to patients’ features). Repositories for sharing data for the development of AI tools could potentially be developed, much like the clinical trial data-sharing repositories that have been developed to date. Although

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challenging, it should be possible to ensure protection of intellectual property rights and confidence of users in AI models. Indeed, this potential for harmony has already been well demonstrated over the course of decades of drug and biomedical device development and approval. We believe that, building on this experience, a reasonable balance may facilitate further AI developments, ensuring reliability of XAI systems for the clinicians who use them.

Regulatory agencies and their AI registration process

AI tools for IBD have potential in many areas, including clinical practice, clinical trials, drug discovery and development, at-home/remote monitoring and beyond. AI models may also help to replace, reduce and refine the use of animal models during the preclinical development of drugs in IBD. In clinical trials, AI systems have the potential to assist with patient selection based on certain disease characteristics or other clinical parameters, while in clinical practice, AI tools could potentially analyse real-world data in real time or near real time to help guide patient management.

One key consideration is that while the majority of tools using AI in the IBD space are currently based on machine learning (ML), CNNs and, increasingly, natural language processing (NLP), these factors can often be wrongfully termed “generative” AI. “Generative AI” refers to generation of new content, whereas ML models learn to use data from previously derived content. Indeed, AI can be applied across the breadth and depth of IBD research and clinical practice, from fundamental laboratory research with analysis of -omics data sets, analysis of electronic health record (EHR) systems and extraction of molecular or digital biomarkers from imaging, endoscopy and histopathology, all the way through to clinical decision support tools and digital

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biomarkers.[55,56] In addition, multiple tools are currently being delivered where AI applications could be incorporated into other novel technologies to assist with monitoring of disease activity, including through the use of smart devices and "wearables".[57]

Regulatory bodies worldwide, including the Food and Drug Administration (FDA) in the USA and the European Medicines Agency (EMA), are currently grappling with the complex task of adapting existing frameworks to accommodate the unique features of AI-based medical devices.[58–60] The FDA, for instance, has embraced a forward-looking approach by engaging in dialogue with industry stakeholders and releasing guidance documents to outline regulatory expectations.[61] Similarly, the EMA has emphasised the importance of ensuring the reliability, robustness and performance of AI algorithms, particularly in the context of clinical decision support systems.[62]

As with medications and medical devices, the key challenge in regulating medical AI for patient care lies in striking a balance between fostering innovation and ensuring patient safety. The iterative nature of AI development, characterised by continuous learning and optimisation, complicates traditional regulatory paradigms built around static hardware devices or medications. Regulatory agencies are thus exploring novel approaches, such as adaptive regulatory pathways and real-world evidence generation, to facilitate the timely and efficient evaluation of AI technologies.

In 2019, the FDA introduced a discussion paper proposing a framework for modifying AI/ML-based medical devices to ensure safety and effectiveness, acknowledging their iterative nature.[63] Subsequently, the FDA developed an AI/ML Software as Medical

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Device Action Plan in January 2021, prioritising the harmonisation of AI/ML best practices, including Good Machine Learning Practice (GMLP).[64] By late 2021, the FDA, Health Canada and the Medicines and Healthcare products Regulatory Agency (MHRA) in the United Kingdom published ten guiding principles for GMLP development.[65]

AI/ML's benefit lies in its ability to optimise performance over time and learn continuously from real-world experiences. However, under traditional FDA regulations, software changes, just like medication “label changes” or technical changes in a medical device, have historically required new risk assessments. To address this, the FDA proposed a Predetermined Change Control Plan (PCCP) in its 2019 Discussion Paper and 2021 Action Plan, enabling manufacturers to predict and implement algorithm changes without additional submissions.[66]

Currently the FDA categorises devices into three classes (I, II and III) based on risk level, with Class III being the highest risk. Class I devices are typically exempt from premarket submissions, while Class II devices may use the 510(k) pathway, demonstrating substantial equivalence to a predicate device. Class III devices require premarket applications (PMA). Clinical decision support software provides healthcare providers and patients with personalised information to enhance healthcare, and in order to be exempt from FDA regulation, clinical decision support software must meet criteria set in the 21st Century Cures Act of 2016 (**Table 1**).[67]

In the European Union (EU), to market a medical product, compliance with either the Medical Device Regulations (MDR)[68] or the In Vitro Diagnostic Regulations (IVDR) is

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mandatory.[66] The EU classifies medical devices into four levels: Class I, Class IIa, Class IIb and Class III, with Class I posing the lowest risk and Class III the highest. The emergence of the EU AI Act represents a watershed moment in AI regulation, signalling a shift towards more comprehensive and harmonised oversight.[69] By mandating stringent requirements for the development, deployment and monitoring of AI systems, the AI Act aims to instil trust and confidence in AI technologies while mitigating potential risks. Key provisions of the AI Act include governance and data management requirements, transparency standards, human oversight mechanisms and cybersecurity protocols. The AI Act was further supported by development, in July 2023, of the Data Privacy Framework as a combined effort by the USA and EU to ensure security of personal data transfers under the General Data Protection Regulation (GDPR) between the EU and the USA.[70]

Impact on the wider life sciences sector

The impact of regulatory decisions extends beyond individual devices to the broader life sciences ecosystem. As AI technologies become increasingly integrated into clinical practice, manufacturers, healthcare providers and patients will need to navigate and be compliant with relatively complex regulatory landscapes in order to mitigate risks and maximise benefit for patients. Additional considerations regarding regulatory decisions will need to consider influence on investment decisions, market dynamics and innovation trajectories, shaping the future landscape of AI in healthcare.[60]

It is important to note that many AI systems may not qualify as medical devices under the AI Act due to lack of medical intent or high-risk designation. This is comparable to

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the current climate for many unregulated wearables, including some for which dubious claims have been made regarding health benefits. Scientific AI applications could also benefit from exceptions under the AI Act, particularly those developed solely for research and development purposes, as indicated by the EU Parliament's focus on supporting innovation. Such pathways are promising, as sub-studies in classic registrational RCTs of medications could be used to evaluate and validate these technologies. Support for such endeavours from professional organisations in the field of IBD could help further facilitate progress in this regard. However, at this time it remains unclear how "scientific research" will be defined and whether this exemption will apply solely to academic or also to commercial research, both of which are often intertwined today.

Current regulatory landscape for AI in IBD

Some clinical decision support software can be exempt from the definition of a device by the FDA but must meet the criteria summarised in **Table 1** (exemptions from FDA review).[66] However, for other AI technologies in IBD, tools that enhance consistency in objective assessment are anticipated to be favourably received by regulatory agencies. This mirrors the adoption of central reading in IBD clinical trials, which significantly improved consistency and reliability, particularly with regard to endoscopic outcome assessment.[71] Although approval of AI tools in IBD is still at a very nascent stage, there has been recent progress, with notable examples providing a potential roadmap for others to follow in the near future (**Table 2**).

In addition to the highlighted IBD-specific examples, it is conceivable that many more existing, mostly non-regulated or non-approved smart devices (e.g. smart watches)

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could be utilised to collect and display AI-relevant data to improve IBD care. Therefore, it is clear that regulatory factors will likely play a major role in influencing the integration of AI technologies into the management of chronic medical conditions like IBD. Accordingly, collaborative efforts among regulators, industry, academia, healthcare providers and patient groups will be crucial in creating flexible and responsive regulatory frameworks. Critically, these frameworks will need to encourage innovation while prioritising patient safety and welfare.

Biomedical AI application submission process

Global Health authorities have made significant progress in harmonising regulatory principles concerning AI in healthcare. This is necessary to nurture initiatives from research and industry alike and to provide regulatory certainty. Documentation is an important aspect of the development process and problems are most likely to arise or be detected when approaching the submission process. Central documents of design control include the Design History File (DHF) and the Device Master Record (DMR) for devices.

The DHF is a documentation that describes design and development activities during the process of developing a medical device (**Table 3**). The aim is to demonstrate that the development and design control process meets the requirements of the FDA (or other regulatory body) as described in FDA Regulation CFR 21 820. Analogous regulatory requirements around the world would be the EU Medical Device Regulation 2017/745 and EN ISO 13485 Quality Management Systems. There are important variations in the validation process in diverse regions of the world that need to be considered, e.g. Annex II 3.1 of EU Medical Device Regulation requires a description

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of the design stages applied to a device to be provided as part of the technical documentation. The described aspects of design control can also be seen as a best practice approach, beyond being a regulatory requirement. It will also be typical for a notified body to audit medical device product use at regular intervals.

The submission process and type, as well as the requirements, are based on risk assessment of the Software as Medical Device (SaMD) system and include 510k notification, De Novo or PMA pathway. Applying the criteria from 21 CFR 807.81 and published guidelines on when to submit a 510(k) for software changes in a medical device, a premarket submission to the FDA is required when the AI/ML software implies a change in its intended use or a major change in algorithms of an existing device or otherwise significantly affects safety, performance and effectiveness. However, whether this approach, derived from the traditional paradigm of medical device regulation, applies to a continuously learning AI algorithm and what metrics may be able to continuously measure performance is an important area requiring focus from regulatory authorities.

As explicitly mentioned in FDA publications,[72] non-device software functions and other functions relating to administrative tasks, maintenance or encouragement of a healthy lifestyle, electronic patient records, the storing, transfer and display of data and the provision of certain limited clinical decision support - are not subject to FDA regulation. The IMDRF SaMD risk categorisation framework is aimed at harmonising regulation.[73] This document is based on risk assessment with the following two major factors: (a) significance of information to healthcare decisions provided by the AI/ML system and (b) the specific condition or situation addressed.

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An important consideration in the submission process is adherence to standards published by standards development organisations, namely the International Organization for Standardization (ISO), the International Electrotechnical Commission (IEC) and the Institute of Electrical and Electronics Engineers (IEEE). These standards are widely recognised by regulatory bodies. IEC and ISO have joined forces to form a technical committee, ISO/IEC JTC1/SC42, which is active in developing standards aimed at AI/ML.[74]

A summary of regulatory steps for the submission process of AI-based software is provided in **Supplementary Table 1**.

The existence of specific funding streams will be important in order to stimulate future academic and commercial research into AI tools for use in IBD. Currently there is a lack of such specific funding streams although there are numerous potential sources of funding, including from governments, private foundations and charities, and industry partnerships (**Supplementary Table 2**). Industry partners need to be identified, taking into account their strategic interest in the field. Important aspects that need to be clarified and agreed at the outset include the ownership of intellectual property rights and the levels of independence of the researchers; therefore, communication and contracting are often key.

Reimbursement for approved and deployed AI applications

Reimbursement is critical to allow translation of any intervention into clinical practice and ultimately patient benefit.[46] With regard to AI tools specifically in the context of

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IBD, there are multiple potential pitfalls and challenges that need to be considered when it comes to reimbursement.

It is undeniable that the direct cost of digital transformation including AI is likely to have a major bearing on the likelihood of reimbursement and adoption in clinical practice. This is complicated by the existence of highly variable reimbursement policies and organisations around the world, with separate healthcare technology agencies (HTAs) for many different countries. The methods used to calculate the costs and consequences for reimbursement of AI tools are so heterogeneous that it would be beyond the scope of this review to cover all possible economic evaluations. An important point to note is that direct (unit) costs, which include the cost of the AI tool itself, may be relatively low. However, low cost of the AI tool itself may not be sufficient. The costs of implementing the AI tool into routine care (pathway costs), including the cost associated with training and educating staff in how to use the tool, may be much higher and may prohibit uptake.

Whilst strictly speaking not directly related to reimbursement of an AI tool, a key indirect consideration is interoperability and potential for integration of an AI tool into routine healthcare services. Most notably, whether an AI tool can be integrated into EHR systems is likely to play a key role when health care providers consider whether a tool is worthy of reimbursement. Accordingly, ensuring integration of an AI tool alongside routine healthcare delivery, with appropriate education and technical support, will likely support reimbursement and adoption in routine care.

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One way to overcome some of the cost uncertainties prohibiting uptake of AI tools in IBD will be to perform RCTs of AI-based interventions—no matter whether the RCT is AI enabled or supported.[75] Indeed, it may well be the case that over time regulatory guidance evolves such that reimbursement of AI tools may be agreed only when there is supportive data from interventional RCTs.[76] Appropriately conducted trials with robust data collection should enable definitive answers to be obtained with regard to efficacy, safety and cost utility—at least over the primary follow-up period for the RCT. A recent example from the IBD field is the PROFILE trial, which assessed a prognostic tool for patients with Crohn's disease, clearly demonstrating its lack of utility in an RCT setting as well as providing helpful information with regard to the optimal management of newly diagnosed patients.[77] A major barrier here might be a requirement for all AI tools in IBD to undergo testing in an RCT prior to implementation in routine healthcare, as this could slow down the translation of AI tools to the benefit of patients. In this regard, a key process which could help speed up the assessment and validation of AI tools would be the assessment of multiple AI tools in parallel within one master trial protocol, so that faster answers could be obtained for patients.[78] Clinical trials using more efficient protocols of this nature have been used with great success across other disease areas and it will be important to implement them in the field of IBD.[79,80]

In the context of AI tools and health economic evaluation, it is still not clear whether the most appropriate means of assessment will be cost-benefit analyses which focus on monetary outcomes, cost-effectiveness analyses which focus on clinical outcomes or cost-utility analyses which focus on quality of life outcomes.[81] Moreover, it is still unclear whether it is appropriate for countries or regions that cannot conduct their own health technology assessments (owing to insufficient resources) to rely on

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assessments conducted in other parts of the world with differing healthcare services and pathways costs. The potential for differential reimbursement around the world poses perhaps the biggest risk of widening inequality and accessibility to AI tools for patients living with IBD. Accordingly, we believe that harmonising reimbursement pathways and processes is an important area of unmet need, which should be urgently addressed by collaborative efforts across academic, regulatory and industry collaborators and partners.

Summary and future outlook

Computational biology and advanced mathematical modelling using AI tools have been applied in the setting of IBD for many years. Indeed, it is important to note that many significant discoveries in IBD have been facilitated by computer science methodology including AI, such as advances in understanding of the genome, microbiome and metabolome. However, recent years have witnessed an increasing ability to handle large electronic health datasets that has been paralleled by a vast increase in computing power and possibilities for bioinformatic analyses. These rapid advances have been accompanied by new generations of patients and healthcare providers who are potentially more open and accustomed to the routine use of digital technologies and AI-generated algorithms/recommendations.

It is only logical to harness this development in our ongoing effort to better understand the causation of IBD and improve individual outcomes for patients. As is true for virtually all new methods and technologies, the potential benefits of AI need to be balanced against the risks and challenges, which need to be addressed before deployment of AI tools in routine clinical care in IBD. Learning from the past requires

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us to acknowledge that AI, just like -omics research, will not answer all the questions we still have about the mechanisms of disease, individual patient disease courses and responses to medications. Importantly, the current hype around AI tools should be tempered with the need for rigorous research to help identify the best possible applications. In particular, it will be important for any potential AI tool in IBD to demonstrate superiority over current human workflows.

Prior to implementation of AI tools in the routine clinical care of IBD patients, it will be important to have clear processes in place to avoid unintended consequences. It will need to be ensured that determination of patient management does not rely solely on AI-derived algorithms. AI tools should remain decision support tools, acting simply as a guide for healthcare professionals and patients. Although the potential risks and consequences of AI are vast, it should also be considered that the potential benefits are perhaps even larger, and we believe that AI tools have the potential to significantly contribute towards the delivery of more effective and safer care in IBD.

In this article, we have provided a roadmap of key considerations in the developmental lifecycle of an AI tool, including the envisioned clinical setting, ethical algorithm design, regulatory submission and registration, potential reimbursement and implementation in routine clinical care. In order for AI to reach its full potential, it will be necessary to harmonise regulatory practices with appropriate methodological rigour. This should ensure successful implementation of AI tools to guide clinical care for patients living with IBD.

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Author contributions

All authors critically reviewed and revised the manuscript, and all authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

Conflict of Interest

NMN reports personal fees from AbbVie, Bristol Myers Squibb, Celltrion, Galapagos, J&J, Lilly, Pharmacosmos, SBK Healthcare and Takeda outside the submitted work and grants from Dr Falk Pharma, Pfizer, Pharmacosmos and Tillotts Pharma AG outside the submitted work. MD reports honoraria for advisory board participation from Abbvie, Janssen, Eli Lilly, Takeda, Pfizer, Galapagos and Alfa Sigma; grants from Eli Lilly and Takeda; consultancy fees from Clario-Bioclinica; and clinical trials with Roche, Eli Lilly, Abbvie and Takeda. JdL reports personal fees/honoraria for advisory board participation from Takeda, Sanofi, Danone and mirum and grants from Takeda, Abbvie, Hexal, Danone and metaX, all outside the submitted work. MA reports consulting fees from Nikkiso Europe, Mundipharma, Janssen, Abbvie, Ferring, Galapagos, Alfasigma and Pfizer. UK reports speaker and advisory fees from Abbvie, BMS, Celtrion, Eli Lilly, Janssen, Medronic, Pfizer, Roche and Takeda and research support from Abbvie, Eli Lilly, Janssen, Medtronic and Takeda. DN reports unrestricted research grants from Pfizer, consulting fees from Abbvie and Dr. Falk Pharma and lecturer fees from Alfasigma and Ferring. TR reports personal fees from Abbvie, Arena, Aslan, AstraZeneca, Boehringer-Ingelheim, BMS, Eli Lilly, Ferring, Galapagos, Gilead, GSK, Heptares, LabGenius, Novartis, Numab, J&J Innovative Medicine, Pfizer, Roche, Takeda, UCB and XAP therapeutics outside the submitted work; grants from AbbVie outside the submitted work; and fees from data monitoring board

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membership of UCB outside the submitted work. SS reports consulting fees from Takeda, AbbVie, Merck, Ferring, Pharmacosmos, Warner Chilcott, J&J Innovative Medicine, Dr Falk Pharma, Biohit, TriGenix, Celgene, and Tillotts Pharma outside the submitted work and payment or honoraria from AbbVie, Takeda, Celltrion, Pfizer, Biogen, AbbVie, J&J Innovative Medicine, Merck, Warner Chilcott and Dr Falk Pharma outside the submitted work. SS is co-director of research for the South Asian IBD Alliance. DCB reports serving on scientific advisory boards and or receiving travel support from AbbVie, Galapagos, Alfasigma, Dr. Falk Pharma, Pfizer, Janssen-Cilag, Takeda and institutional research grants from Janssen Cilag and Pfizer, all outside the submitted work.

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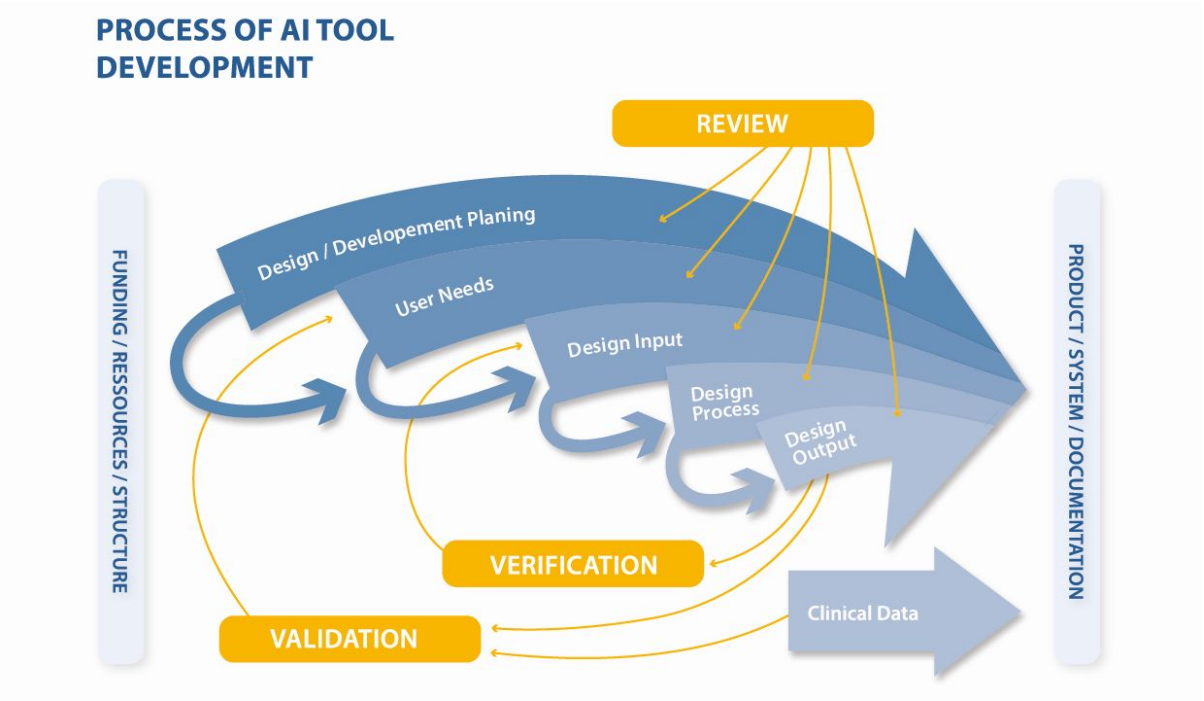
Figure Legend

Figure 1. Process of AI tool development. An overview of the main steps involved in taking an AI tool from development through the regulatory submission process to clinical application.

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Figures

Figure 1.



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948 **Table 1. Excerpts from "Exemptions from FDA review".[67]**

Congress excluded certain health-related software from the definition of a medical device in the 21st Century Cures Act of 2016.55. Exemptions in the law include software that: • Is intended for administrative support of a health care facility. o Example: software for scheduling, practice and inventory management. • Is intended for maintaining or encouraging a healthy lifestyle and is unrelated to the diagnosis, cure, mitigation, prevention, or treatment of a disease or condition. o Example: mobile applications that actively monitor exercise • Is intended to serve as electronic patient records, including patient-provided information, and its function is not intended to interpret or analyze patient records for the purpose of diagnosis, cure, mitigation, prevention, or treatment.

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Table 2. Examples of AI applications in IBD.

Regulatory approval in place or pending:
• **"EndoBRAIN – UC"** (Olympus/Cybernet), which was granted regulatory approval in Japan in April 2020 and launched in February 2021 (only in Japan).[82] • **"Red Density"** (Pentax Medical)—which has been approved in Europe and is expected to be commercially available in Europe later in 2024.[83]
Commercially available in the area of bio-discovery and research and development for the pharmaceutical industry where there is no direct regulatory requirement:
• **"Certai"** (Satisfai Health), an AI-powered scoring tool for the assessment of ulcerative colitis in stage I through IV clinical trials.[84]
Not commercially available but available under approved clinical study investigational device use conditions:
• **"IBD aware"** (EnLiSense CCM). This device, placed on a patient's skin, collects nanolitre samples of passive sweat, continuously measuring and reporting biomarkers. It includes an AI-powered dashboard, providing real-time actionable insights to the patient.[85]

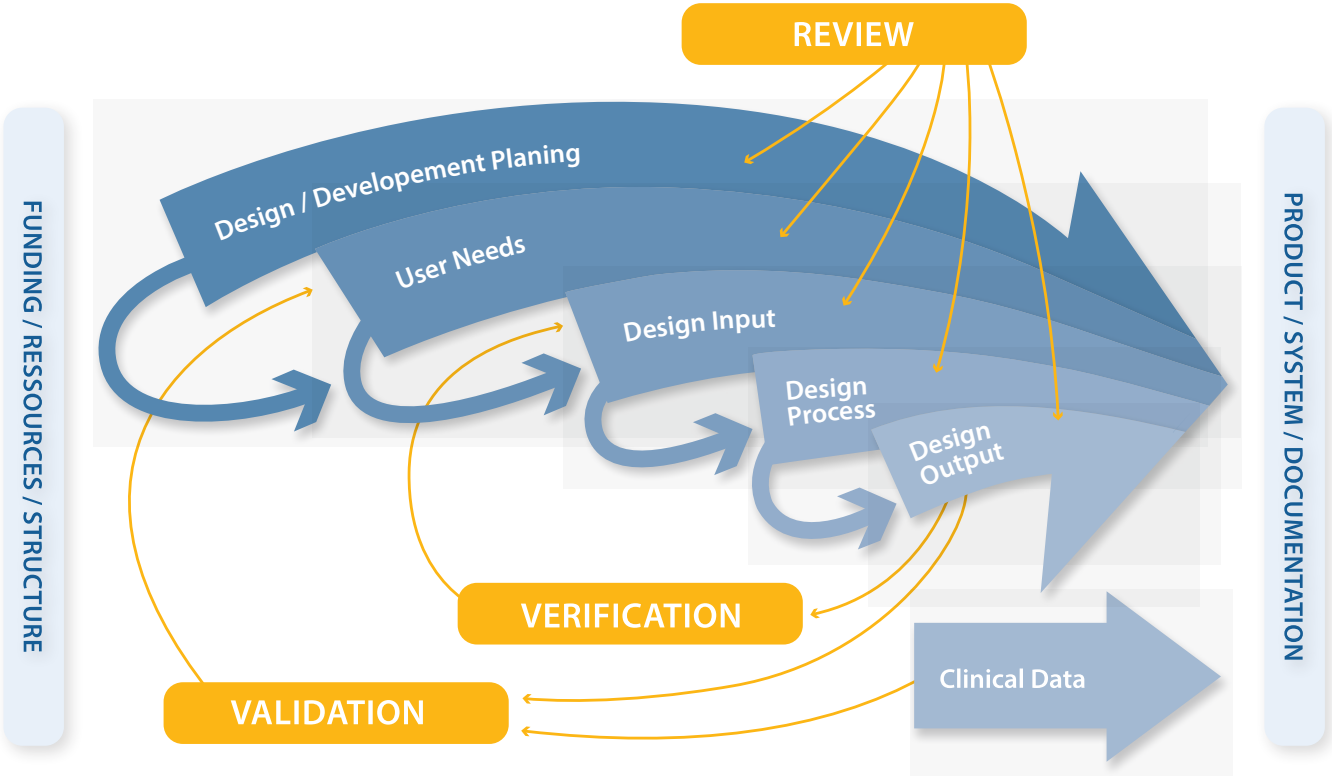
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Table 3. Typical components of the DHF.

1. Design and development planning
2. Intended use and user needs
3. Design inputs and outputs
4. Design verification protocol and reports
5. Design verification demonstrating that the design output meets the design inputs and requirements
6. Design validation protocols and reports
7. Process validation and work instructions
8. Structures and processes setup for productive use
9. Design review meeting notes
10. Design requirement matrix
11. Risk management file
12. Clinical data summary/evaluation report
13. Approval

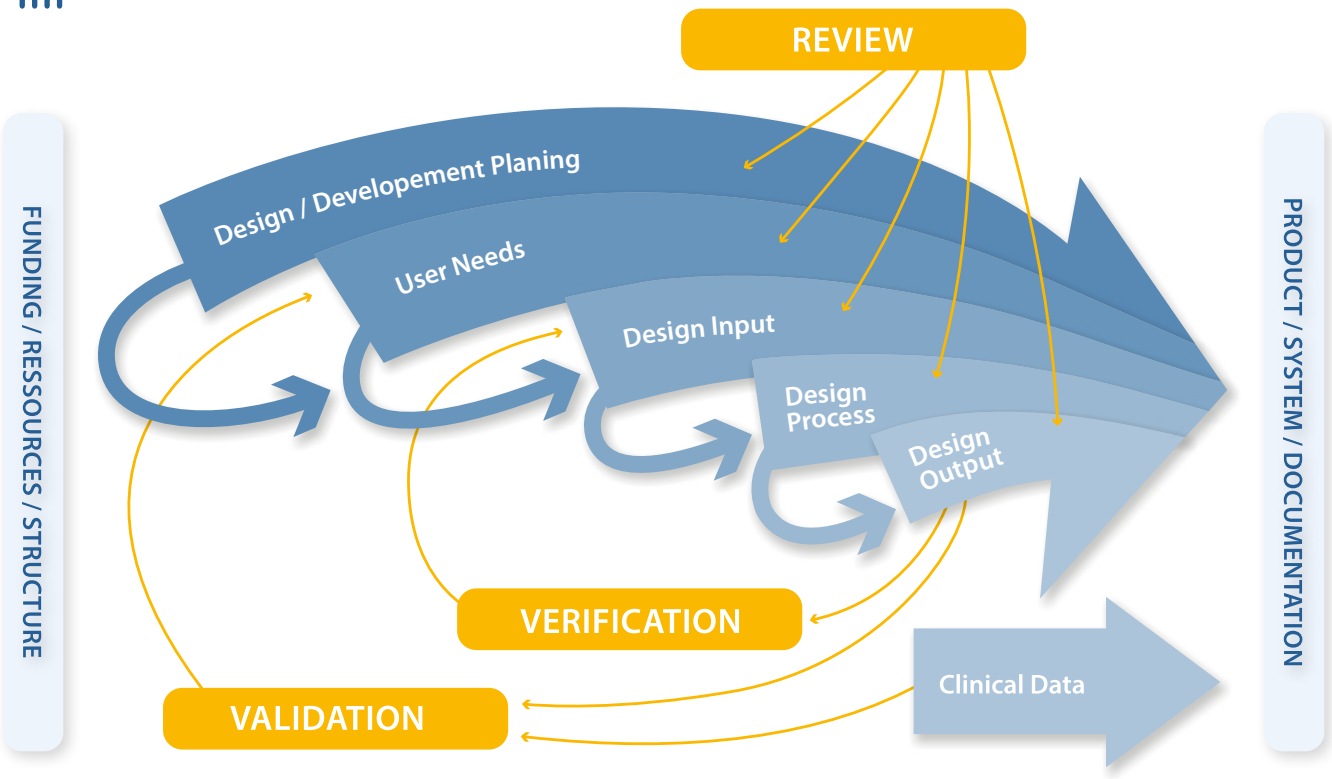
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PROCESS OF AI TOOL DEVELOPMENT



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PROCESS OF AI TOOL DEVELOPMENT



Computational biology and use of advanced mathematical modelling including AI tools is increasingly used in IBD research.

Rapid technological advances have been accompanied by new generations of patients and healthcare providers (, potentially more open and accustomed to the routine use of digital technologies and AI-generated algorithms/recommendations.)

Potential benefits for AI need to be balanced with the risks and challenges, which need to be addressed before deployment into routine clinical care in IBD.

AI just like -omics research will not answer all the questions we still have about the mechanisms of disease, individual patient disease courses and response to medications.

Roadmap of key considerations in the developmental lifecycle of an AI tool including the envisioned clinical setting, ethical algorithm design, regulatory submission and registration, through to potential reimbursement and implementation into routine clinical care.

In order for AI to reach its full potential, it will be necessary to harmonise regulatory practices, with appropriate methodological rigour.

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